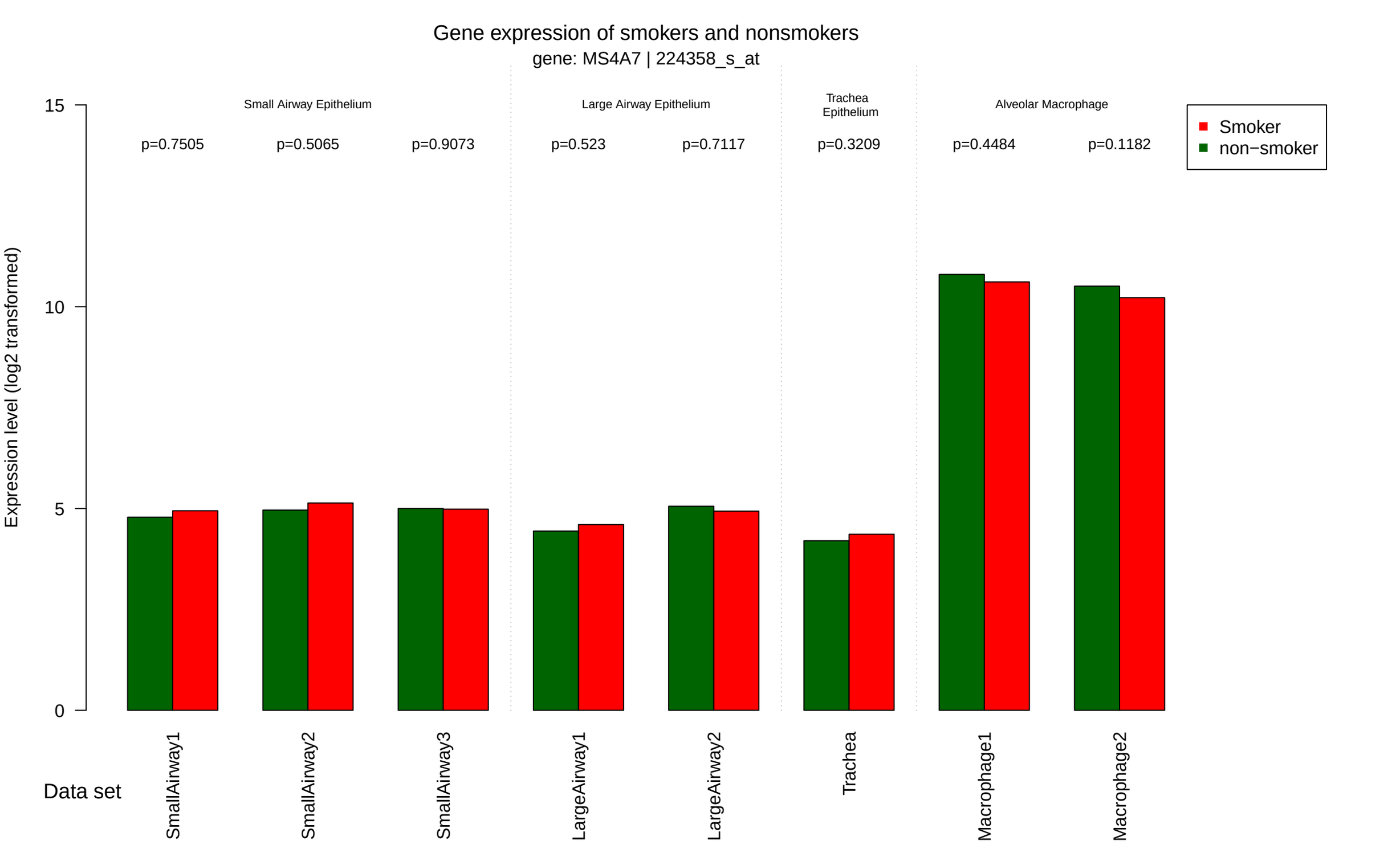
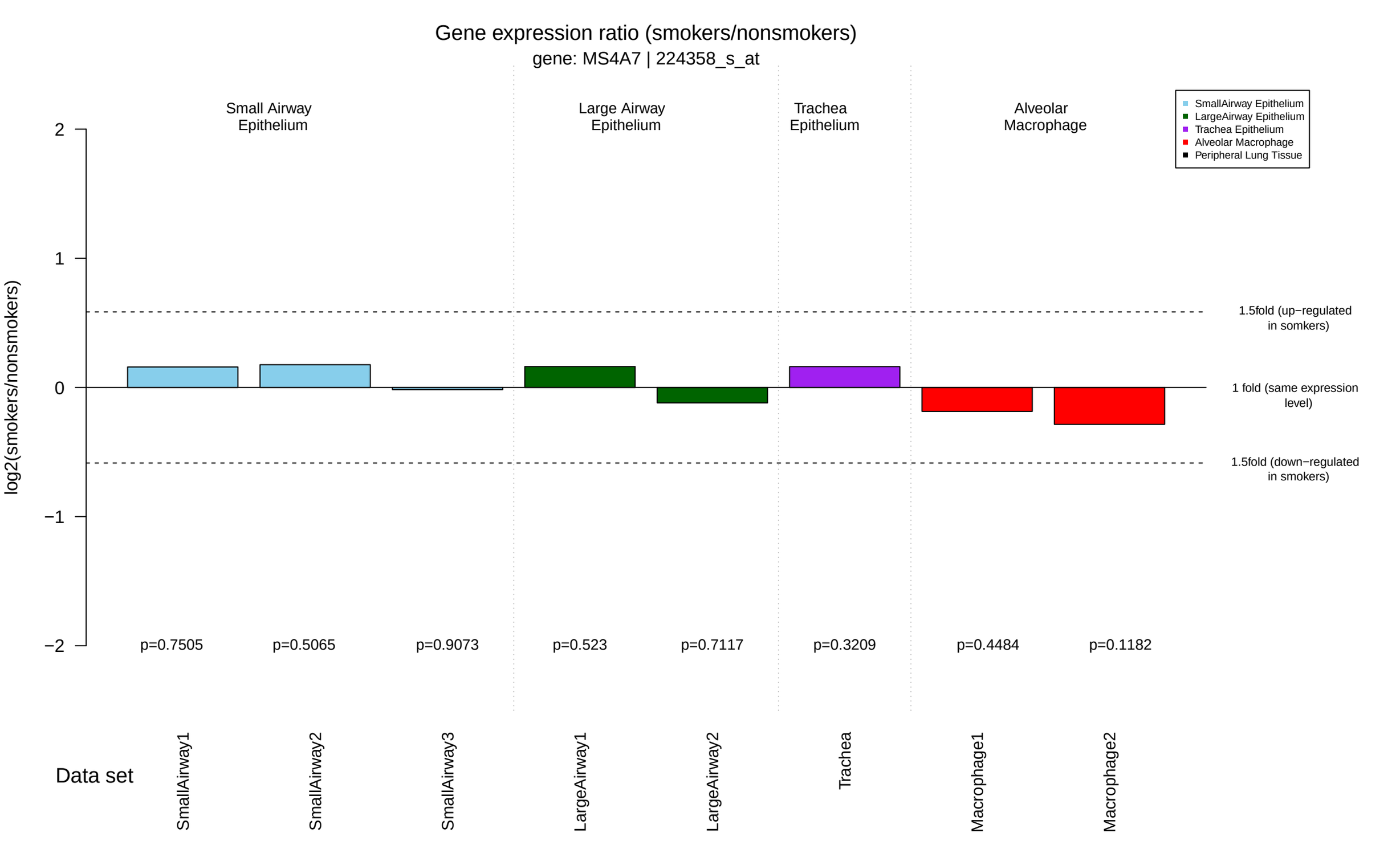
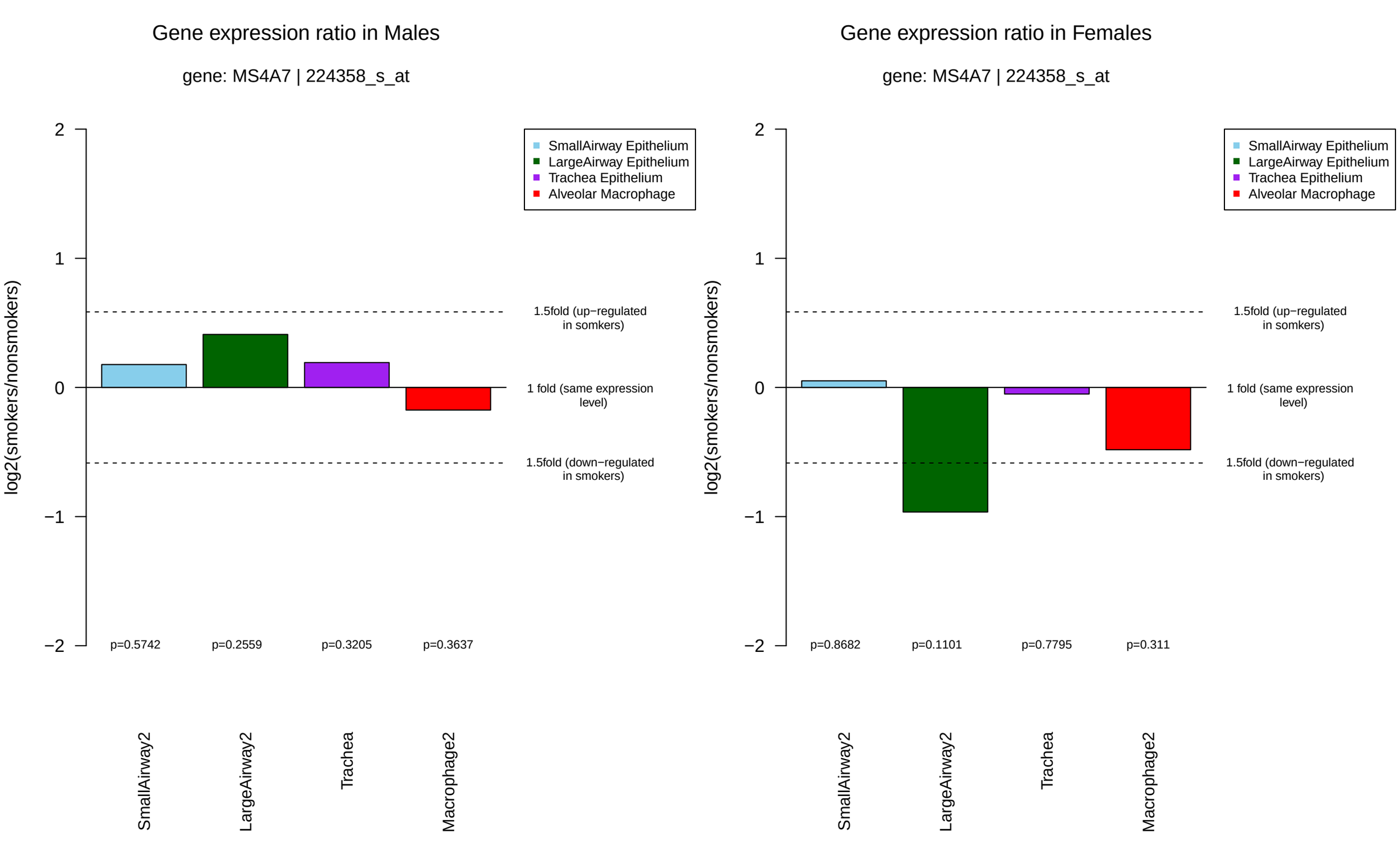
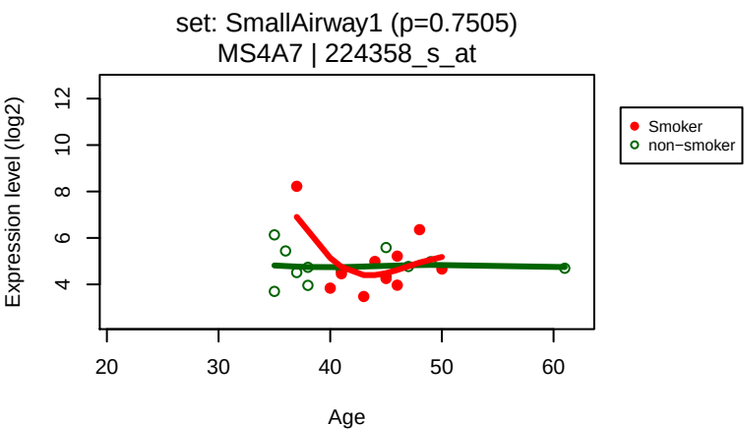
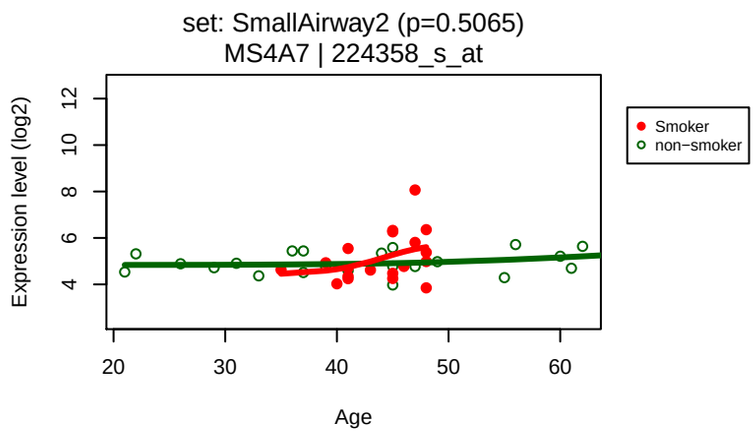
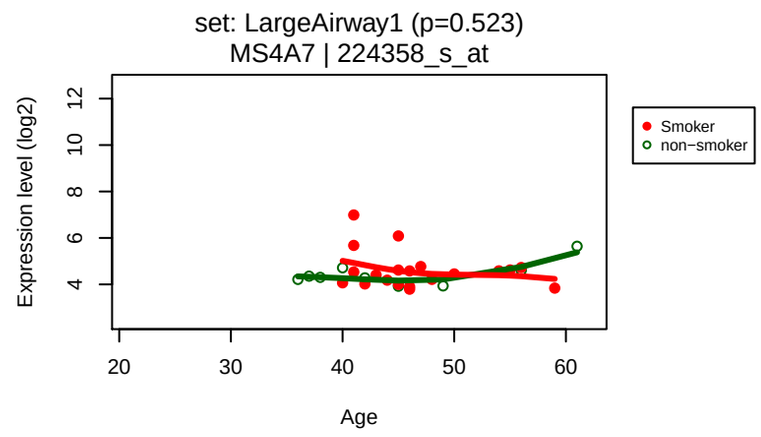

Gene expression of smokers and nonsmokers
gene: MS4A7 | 224358_s_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
p=0.7505
p=0.5065
p=0.9073
p=0.523
p=0.7117
p=0.3209
p=0.4484
p=0.1182
Smoker
non−smoker
Expression level (log2 transformed)
15
10
5
0
Data set
SmallAirway1
SmallAirway2
SmallAirway3
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2

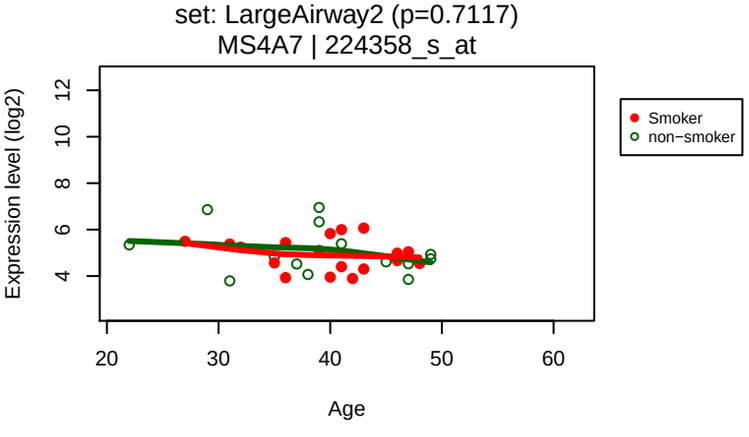

Gene expression ratio (smokers/nonsmokers)
gene: MS4A7 | 224358_s_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
Peripheral Lung Tissue
log2(smokers/nonsmokers)
2
1
0
−1
−2
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.7505
p=0.5065
p=0.9073
p=0.523
p=0.7117
p=0.3209
p=0.4484
p=0.1182
Data set
SmallAirway1
SmallAirway2
SmallAirway3
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2

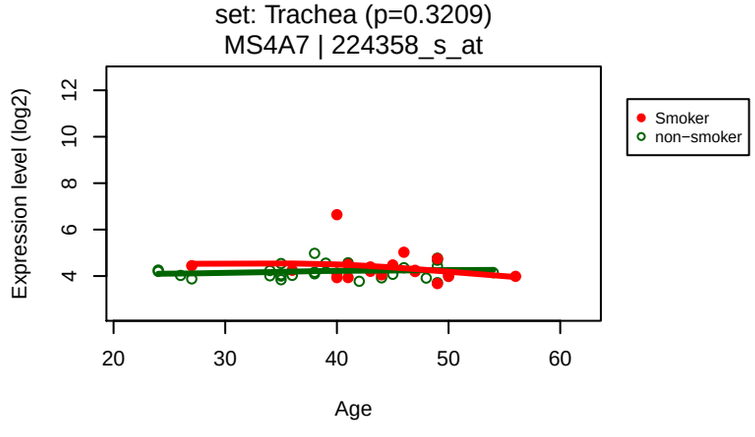

Gene expression ratio in Males
gene: MS4A7 | 224358_s_at
log2(smokers/nonsmokers)
2
1
0
−1
−2
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.5742
p=0.2559
p=0.3205
p=0.3637
SmallAirway2
LargeAirway2
Trachea
Macrophage2
Gene expression ratio in Females
gene: MS4A7 | 224358_s_at
log2(smokers/nonsmokers)
2
1
0
−1
−2
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.8682
p=0.1101
p=0.7795
p=0.311
SmallAirway2
LargeAirway2
Trachea
Macrophage2

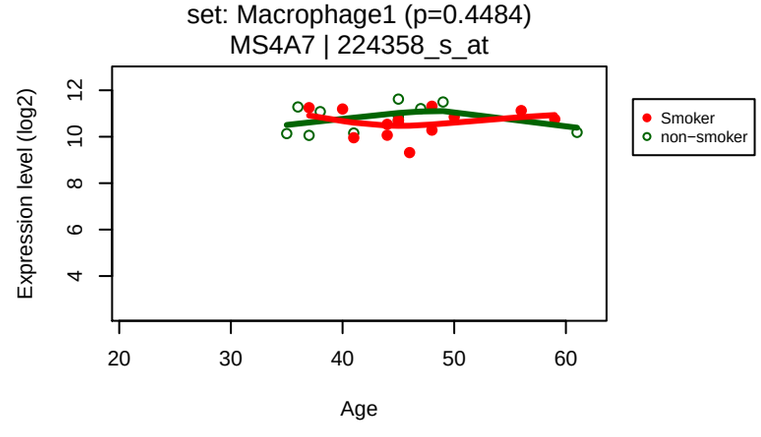

set: SmallAirway1 (p=0.7505)
MS4A7 | 224358_s_at
Expression level (log2)
Age
Smoker
non−smoker

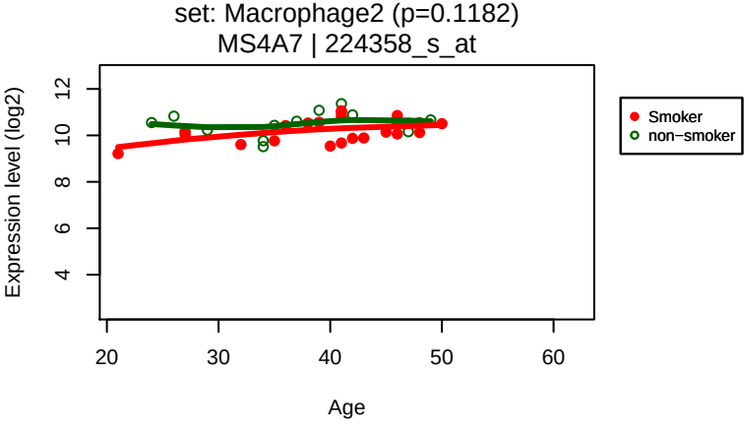

set: SmallAirway2 (p=0.5065)
MS4A7 | 224358_s_at
Expression level (log2)
Age
Smoker
non−smoker

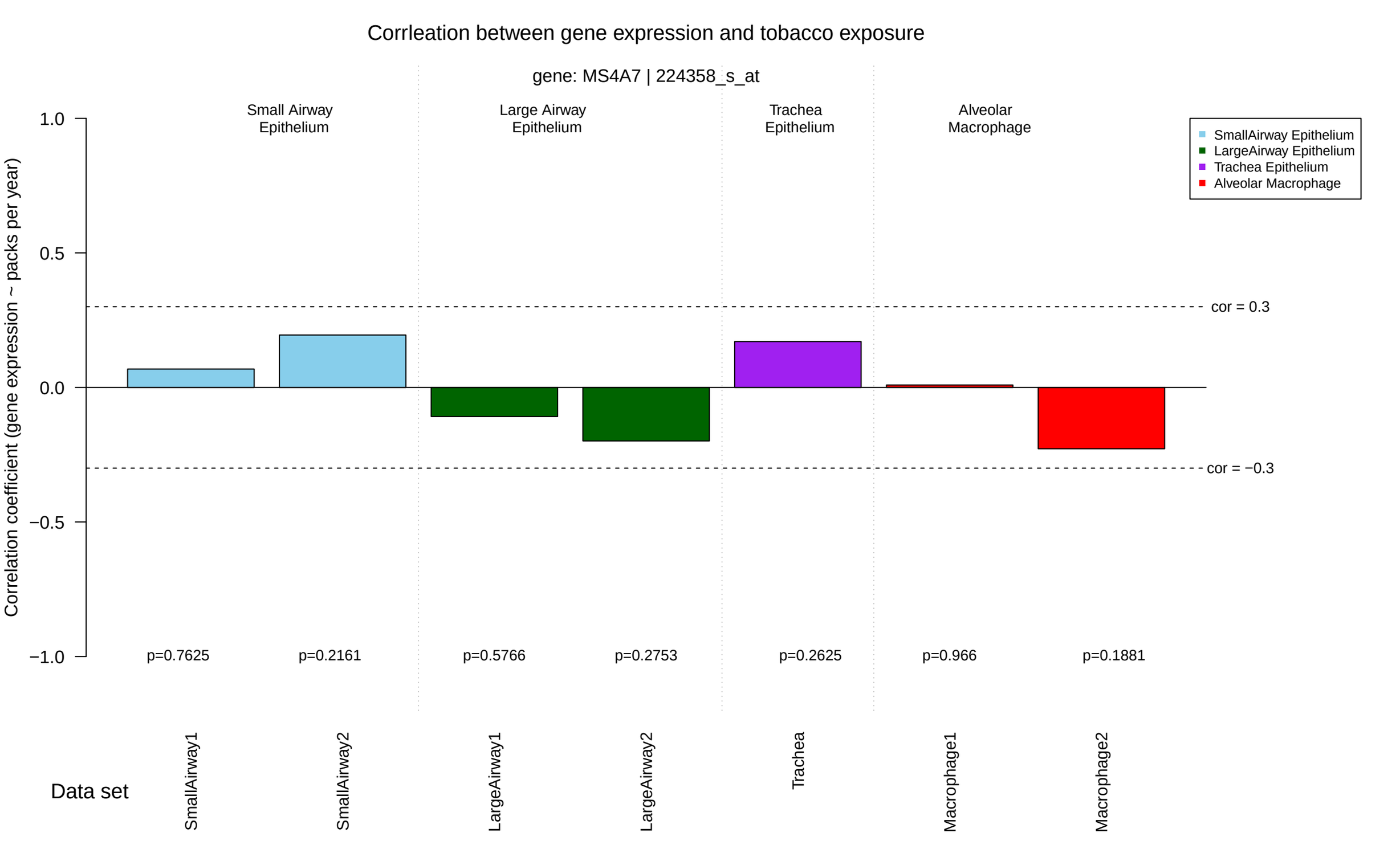

set: LargeAirway1 (p=0.523)
MS4A7 | 224358_s_at
Expression level (log2)
Age
Smoker
non−smoker


set: LargeAirway2 (p=0.7117)
MS4A7 | 224358_s_at
Expression level (log2)
Age
Smoker
non−smoker


set: Trachea (p=0.3209)
MS4A7 | 224358_s_at
Expression level (log2)
Age
Smoker
non−smoker


set: Macrophage1 (p=0.4484)
MS4A7 | 224358_s_at
Expression level (log2)
Age
Smoker
non−smoker


set: Macrophage2 (p=0.1182)
MS4A7 | 224358_s_at
Expression level (log2)
Age
Smoker
non−smoker


Corrleation between gene expression and tobacco exposure
gene: MS4A7 | 224358_s_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
Correlation coefficient (gene expression ~ packs per year)
1.0
0.5
0.0
−0.5
−1.0
cor = 0.3
cor = −0.3
p=0.7625
p=0.2161
p=0.5766
p=0.2753
p=0.2625
p=0.966
p=0.1881
Data set
SmallAirway1
SmallAirway2
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2